HIGH-POWER AND HIGH-EFFICIENCY ELECTROSTATIC ACTUATOR

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ABSTRACT

The authors have previously developed two types of concatenated electrostatic actuators which are driven by pulse excitation profiles. One of these actuators had excellent performance comparable to that of electromagnetic motors. However, the efficiency of the driving circuit for the actuator was insufficient for practical applications. To improve the efficiency, a new type of electrostatic actuator that can be driven by an ac excitation profile was developed. The new actuator is a linear surface-drive actuator weighing 7g and consisting of a thin slider and a stator. Each element has 3-phase electrodes with the same pitch and 3-phase ac driver connected to the electrodes operate the actuator. A new electrode structure and operation method improve the efficiency by a factor of approximately four. In addition, force and power generation performances are improved. It produces a large power of 1.6W and a propulsive force of 4.4N. The power / weight ratio is improved by a factor of about three, reaching top values of 230W/kg.

INTRODUCTION

Up to now, most electro mechanical systems like those found in industrial robots have employed electromagnetic motors such as dc servo motors because of their ease-of-fabrication in the scale of the applications, large power outputs and high controllability. However, these motors are no longer efficient sources of mechanical power, as the size of overall systems and their mechanisms become small and complicated. Such systems need compact and lightweight actuators, but it is difficult to reduce the weights of these motors since they need high-density magnetic materials. Alternatively, electrostatic actuators need no heavy materials and are easily fabricated into lightweight, compact units. Yet, in the past,it was been believed that electrostatic actuators could not produce as much power as electromagnetic ones and, as a result, there have been few applications of electrostatic actuators. Recently, an electrostatic actuator with a size of 1mm or smaller was considered to be capable of producing more power and force than an electromagnetic one with the same dimensions, and various micro-sized motors were developed in the past few years [1,2,3]. Techniques of these motors can be applied not only to micromachines but also macroscopic machines when a large number of micro actuator units are concatenated and their output forces are summed together [2,4,5].

The most appropriate configuration for a concatenated actuator unit is that of linear surface-drive [4]. It generates force between the entire facing surfaces of its slider and stator. The volume of the slider and the stator is easily reduced by cutting down the thicknesses of the slider and stator without any decrease of the force generation. When many actuator units are located in series, the output forces can be combined easily. Concretely speaking, their concatenation is done by increasing of the length and the number of the electrodes. Such an actuator is easily fabricated using photolithography and etching techniques or printing. In addition, the output of the concatenated actuator can be multiplied still further by stacking the actuators into a number of parallel layers.

We have formerly developed a *film actuator* [4] and *variable capacitance motor with active slider* [5]. The film actuator consists of a stator with 3-phase electrodes and slightly conductive polymer film. It has the advantage of easy fabrication. Each phase of electrodes was excited with a

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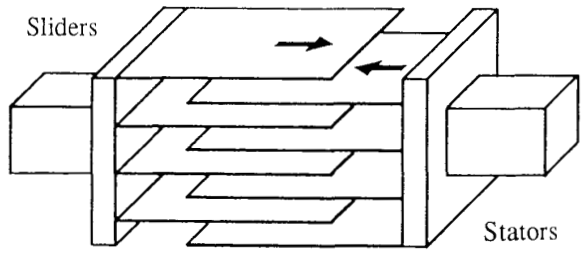


Figure 1 Schematic figure of stacked actuator. Each sheet is a concatenated actuator with a number of electrodes. The outputs of the actuator are multiplied again, by stacking them into layers.

defined pulse profile. The top value of its force density was $2N/m^2$, and its multi-layered type of five layers generated 0.7N of propulsive force. The variable capacitance motor consisted of a stator with 3-phase electrodes and a slider with 4-phase ones. The phases totaling seven were also excited independently with pulse profiles. As for fabrication, this actuator was less advantageous than the film actuator, but it had better performance of force and power generation. The actuator produced 1N of propulsive force. A ten actuator stack produced 21N of propulsive force and a 0.55W of mechanical power. When the actuator was equipped with an electrostatic position sensor and embedded in a closed-loop control system, its power performance improved by a factor of about two, reaching top value of 1.2W mechanical power. The weight of the multi-layered type was 70g, and the power density was 77W/kg. These performance figures are comparable to those of any electromagnetic motors. However, the film and variable capacitance actuators had problems with the efficiency of their driving circuits. For the driving circuits, the actuators are seen as capacitors and, for the most part, the electric energy that the circuits supply is reactive. Generally speaking, it is difficult not to waste the reactive power when a pulse excitation profile is applied to a capacitor. If the profile is sinusoidal, it is easy to recycle the reactive power for devices such as *LC* resonator or inverter circuit. We propose a novel linear surface-drive actuator which can be driven by ac excitation profile.

OPERATING METHODS AND THEIR PRINCIPLES

The actuator consists of two elements: a stator film and a slider film. Three phases of electrodes with the same pitch are fabricated in the both elements. There are three different methods that can drive this actuator: ac-dc operation, dual frequency operation, and single frequency operation.

Ac-dc operation employs a 3-phase ac source and two dc power supplies as shown in Figure 2. Three voltages, positive, negative, and zero are applied to the three phases of the slider's electrodes and charges are induced into the electrodes as shown in Figure 2. The 3-phase ac voltage applied to the stator's electrodes induces a traveling field on the stator. This wave acts against the charges and drives the slider. An experiment for this method was successful, but the operation does not last long due to imperfect insulation of the material of the slider. When constant voltages are applied to electrodes of the slider, charges are induced in the insulation around the electrodes to counter those charged on electrodes, diminishing the field interaction.

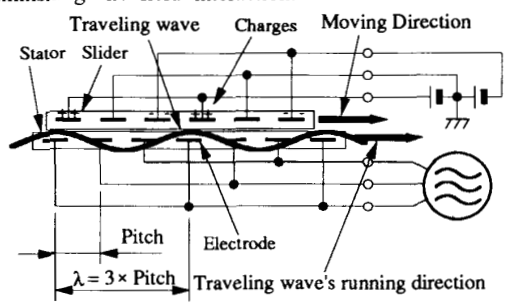


Figure 2 Schematic figure for ac-dc operation.

Dual frequency operation employs two 3-phase ac power sources with different frequencies. The three phase electrodes of both the stator and the slider are connected to 3-phase ac supplies as displayed in Figure 3. Since alternating voltages are applied to all the electrodes, no offsetting charge-up of surrounding insulation occurs. The two 3-phase ac excitations induce traveling fields on the stator and the slider as depicted in Figure 4. Since the wavelengths of the waves are the same, the phase difference δ between the waves can be defined. When the difference is small, field interaction between the two waves generates a repulsive force. The normal component of the force reduces the friction between the slider and stator, and the tangential component drives the slider. The tangential component is dependent on the phase difference sinusoidally, as shown in Figure 5. The velocities of the travelling waves on the stator and the slider, u_1 and $\hat{u}_2$ are

$$u_1 = \frac{\omega_1}{2\pi}\lambda \qquad (1)$$
$$\hat{u}_2 = \frac{\omega_2}{2\pi}\lambda$$

where the coordinate systems of u_1 and $\hat{u}_2$ are fixed to the stator and slider, respectively. When the operation is synchronized under constant load, the phase difference is preserved. The velocities of the traveling waves are the same on the absolute coordinate system. Therefore, the synchronized velocity of the slider u_a is the velocity of the coordinate

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system of $\hat{u}_2$,

$$u_a = u_1 - \hat{u}_2 = \frac{\omega_1 - \omega_2}{2\pi}\lambda \qquad (2)$$

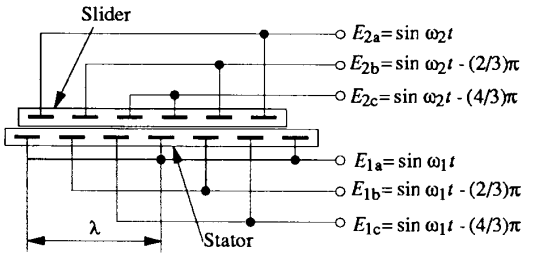


Figure 3 Schematic figure for dual frequency operation.

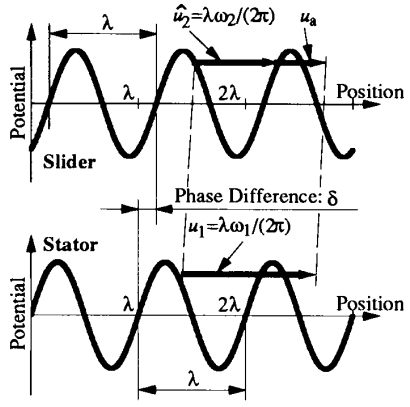


Figure 4 Traveling waves on the stator and slider. When the operation is synchronized, u_1 equals to $\hat{u}_2$ and u_a

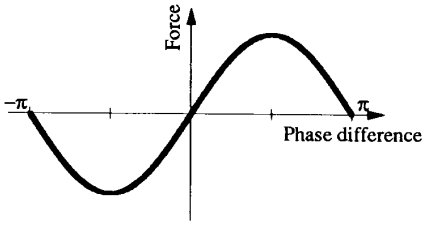


Figure 5 The relation of the tangential component of the force and phase difference between the two waves is sinusoidal.

Figure 6 shows single frequency method, which employs a single 3-phase ac power source. The electrodes of the stator and the slider are excited by the same 3-phase ac, but the orders of connection are inverted. In this case, two travelling waves running in the opposite directions to each other are induced. The velocities of the traveling waves on the stator and the slider, u_1 and $\hat{u}_2$, and the synchronized velocity of the slider, u_a are

$$u_1 = \frac{\omega}{2\pi}\lambda; \qquad \hat{u}_2 = -\frac{\omega}{2\pi}\lambda \qquad (3)$$

$$u_a = u_1 - \hat{u}_2 = \frac{\omega}{\pi}\lambda$$

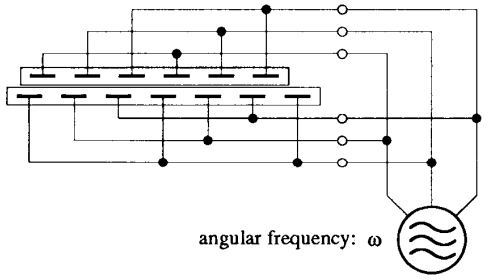


Figure 6 The single frequency operation employs an ac power supply. The electrode connection order on the slider and stator are inverted to each other.

MERITS

This actuator has three main merits below:

1. Linear surface-drive configuration. A single unit of this actuator has three electrodes in the slider and the same number of electrodes in the stator. Configuration of linear surface-drive allows each unit to maintain a large area where electrostatic force is generated. In addition, the units are easily concatenated by increasing the number of electrodes and expanding the length of them. To realize this, photo-lithography and etching or printing techniques are available.

2. Repulsive force drive. This actuator produces large force because of large effective area. Normally, a large contact area increases frictional force, but this actuator is free from the problem of friction since the repulsive force which drives the slider reduces the friction as it drives this actuator.

3. Ac excitation drive. A 3-phase ac power supply is employed by the single frequency method. The power supply can be obtained easily since an inverter circuit, an LC resonator or even a commercial 3-phase power outlet can drive the actuator. In addition, the efficiencies of each of these are quite high.

FABRICATION

The stator and slider have the same structure. In each element, three blocks of electrodes are fabricated as shown in Figure 7. Each block has 312 lines of electrodes with pitch of 320μm. The blocks on the both sides will be used as an electrostatic position sensor in the future, but in this report, all the blocks of the electrodes work for actuation. Figure 8 is the cross sectional view of the actuator. The electrodes are processed by etching copper film away from polyimide base film. The electrodes are covered with polyimide film. Dielec-

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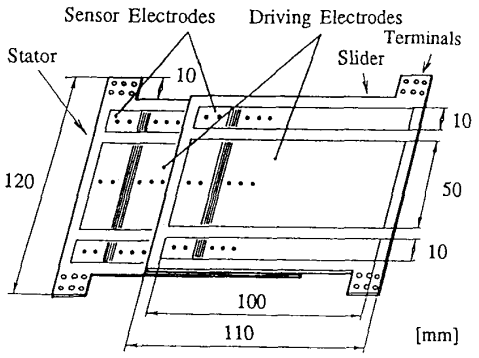


Figure 7 Layout of the electrodes. Sensor electrodes have the same structure as driving electrodes and are used for actuation in this report.

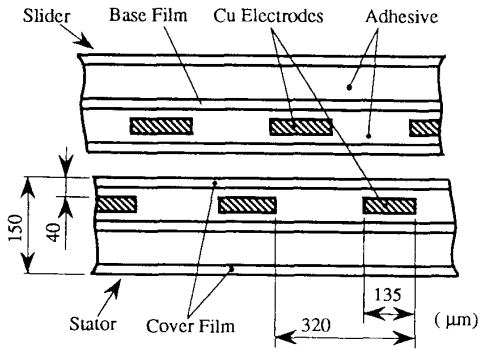


Figure 8 Cross sectional view of the actuator.

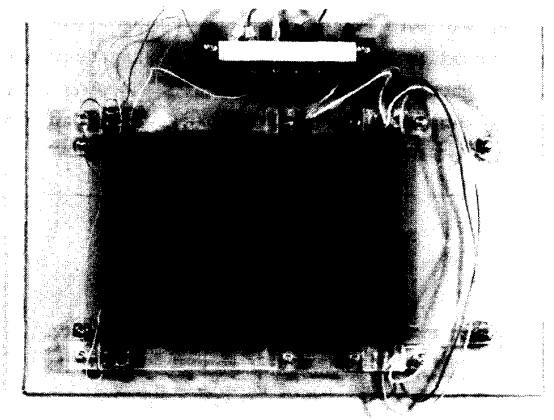


Figure 9 Photo for experimental set up of the actuator. Cubes of acryl resin 47mm long are fixed on the both sides of the slider to keep it from yawing. Two plates 10mm wide prevent the slider from curling.

tric fluid, "Fluorinert (3M)", is injected between the stator and the slider to reinforce the dielectric strength. The total weight of the stator, the slider and the fluid is approximately 7g. Figure 9 is a photo of the experimental set up of the actuator. Narrow plates of acryl resin is set upon the slider to prevent the element from curling, keeping the slider close to the stator. Cubes on the both side of the slider, guide the movement of the slider.

EXPERIMENTAL RESULTS

Comparison of single frequency method and dual frequency method As described previously, three methods can operate the actuator. The first employs two dc power supplies and a 3-phase ac power source; the second, employs ac supplies; and the third, employs a single ac supply. Figure 10 compares the force generation of the last two methods. No difference is observed between the two methods.

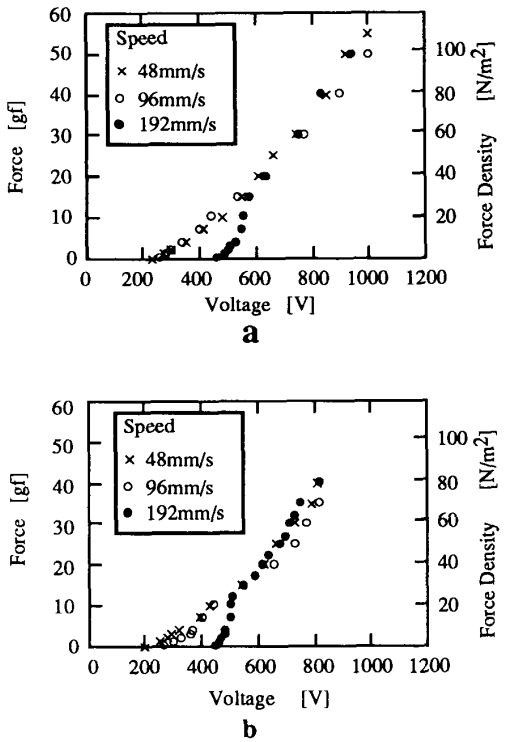


Figure 10 Force and excitation voltages of the actuator. **a.** Single frequency method. **b.** Dual frequencies method.

Movement of the slider Figure 11 depicts the movement of the slider. Synchronized operations where the slider moves at the speeds, 0.78mm/s, 96mm/s and 192mm/s are observed in the diagram **a, b** and **c**. Fluctuation of the movement is due to torque ripples.

Effect of gap The force generation of the actuator depends on the gap between the electrodes of the slider and the stator. Figure 12 shows relations of force generation and excitation voltage when the gap is varied. The actuator generate a force which is basically proportional to the square of the excitation voltage. A large force is produced when the gap is small. When the gap is small and the voltage is high,

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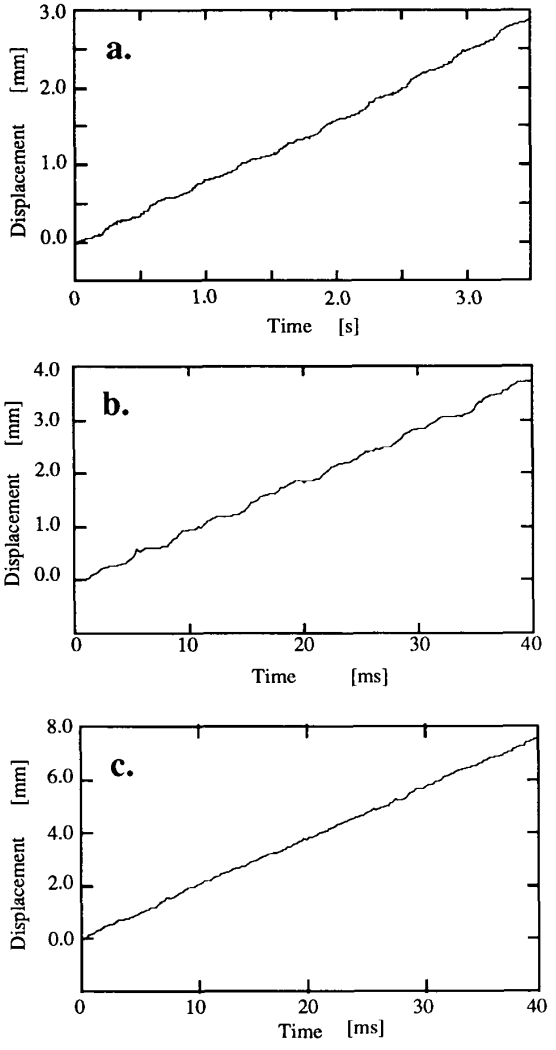


Figure 11 Movement of the slider driven by the single frequency method. The frequencies of the 3-phase acs are **a.** 0.78mm/s, **b.** 96mm/s, **c.** 192mm/s.

the force increases linearly to the excitation voltage. This is due to breakdown which occurs inside the gap.

Effect of dielectric fluid The dielectric fluid injected into the gap improves dielectric strength between the stator's and the slider's electrodes. Figure 13 shows force generation of this actuator using various materials injected. When no fluid is injected, voltage higher than 600V causes breakdown. A small effect of the permittivity of fluid is observed. When silicon oil is injected, the actuator produce its maximum propulsive force 4.4N.

Dynamic characteristic of the actuator Figure 14 shows the force and power generation of operating speed. The actuator produce 1.4m/s linear motion under no load. The power generation reaches the top value of 1.6W at the speed of 0.63m/s.

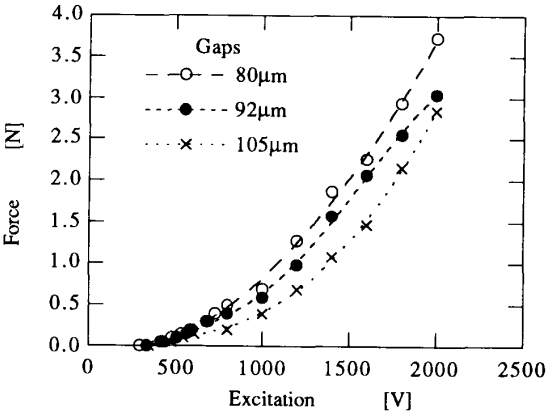


Figure 12 Force generations of the actuator with various gaps between the stator's electrodes and the slider's ones.

Efficiency of power conversion Figure 15 shows excitation profile of one phase of the power source and current it supplies. The phase difference between two diagrams, which is close to 90°, shows that most of the current supplied by the power source is reactive. However, the reactive power can be stored in a device such as reactor. Approximately 22% of the active component is available as mechanical power output. The other energy is consumed by viscosity of the dielectric fluid.

Comparison with other actuators Table I shows the performances figures of this actuator exceeding those of other electromagnetic actuators. In particular, its power / weight ratio is even comparable that of the muscle of natural creatures.

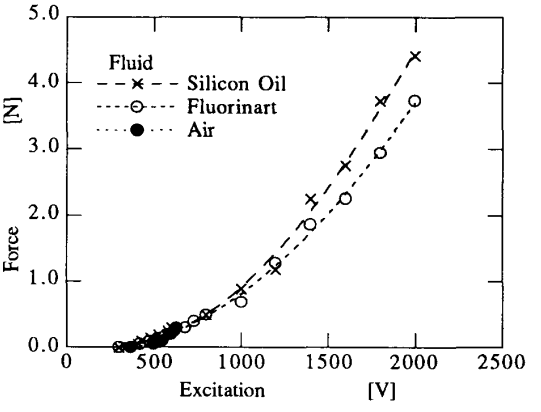


Figure 13 Force generation of the actuator using various dielectric fluid. The perimittivities of Fluorinart and silicon oil are $2.5\varepsilon_0$ and $1.9\varepsilon_0$.

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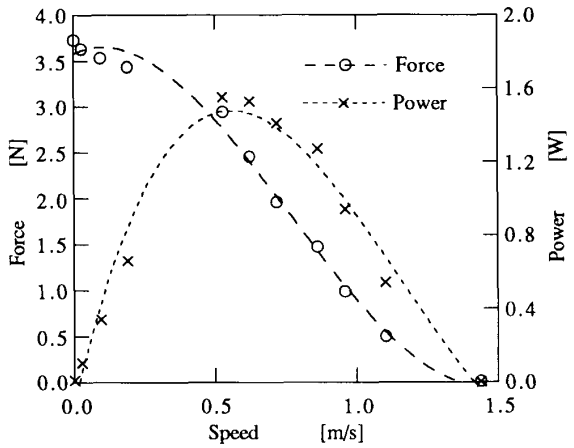


Figure 14 Dynamic characteristic of the actuator.

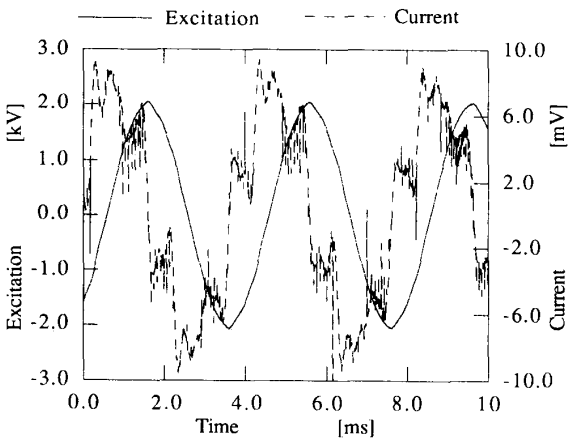


Figure 15 Voltage and current supplied by a phase of the power source, when the actuator pulling up a weight of 200g at a speed of 48cm/s.

Table I Performances of the actuator. **A.** The electrostatic actuator **B.** a voice coil motor (LV01: Sumitomo Special Metals Co., LTD.) **C.** a DC motor (DN26: Canon.)

	A	B	C
power [W]	1.6		1.2
propulsive force [N]	4.4	1.0	
operation speed [m/s]	1.4		
weight [g]	7.0	31*	60
power / weight [W/kg]	230		20
force / weight [N/kg]	630	32	

* The weight of only rotor.

CONCLUSION

We have proposed an electrostatic actuator driven by a 3-phase ac power supply. The actuator was developed for the purpose of improving the performance of power conversion efficiency of the electrostatic actuator formerly developed. To realize this, ac has been selected as the excitation profile of the electrodes, and three types of operating methods (ac-dc, dual frequency and single frequency operation) have been proposed. Among these operation methods, single frequency is most efficient for practical applications since it can be operated for a long duration and the operating apparatus is most simple.

We have confirmed through experiments the following:
(i) An operation using sinusoidal excitation profile is successful. The new structure of the electrodes and the operating method of the actuator improve the efficiency of power conversion by a factor of about four, reaching 22%.
(ii) In addition to removing the problem of efficiency, the actuator has better performances. Power / weight ratio and force / weight ratio is improved by a factor of three and 3.5, respectively.

Since parameters of the actuator are not yet optimized, better performances will be achieved by modifying such parameters as thickness of the slider or stator, the gap between electrodes, their pitch and width. The operation speed and power generation will be improved by a feedback control method using the position sensor.

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